

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0738

Scenic Woods 100 kV
March 01, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0738. The Interconnection Customer proposes a 66.8 MW solar photovoltaic facility interconnecting at the Scenic Woods 100 kV substation in the Indiana area. The request was received on February 01, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 226 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$211,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Minnhaha – Napoleon Peaking Station 230kV Line	Richland – Troy Energy Llc 500kV	104.6	563	12.1
Montpelier Electric Generating Station 500/230 kV Transformer Bank	Montpelier Electric Generating Station – Wabash 500kV L	100.4	804	7.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Minnhaha – Napoleon Peaking Station 230kV Line	Reconductor line	\$12,044,000	1% (\$79,000)
Montpelier Electric Generating Station 500/230 kV Transformer Bank	Replace transformer with higher-rated unit	\$35,569,000	0% (\$132,000)
	Total allocated to project		\$211,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.